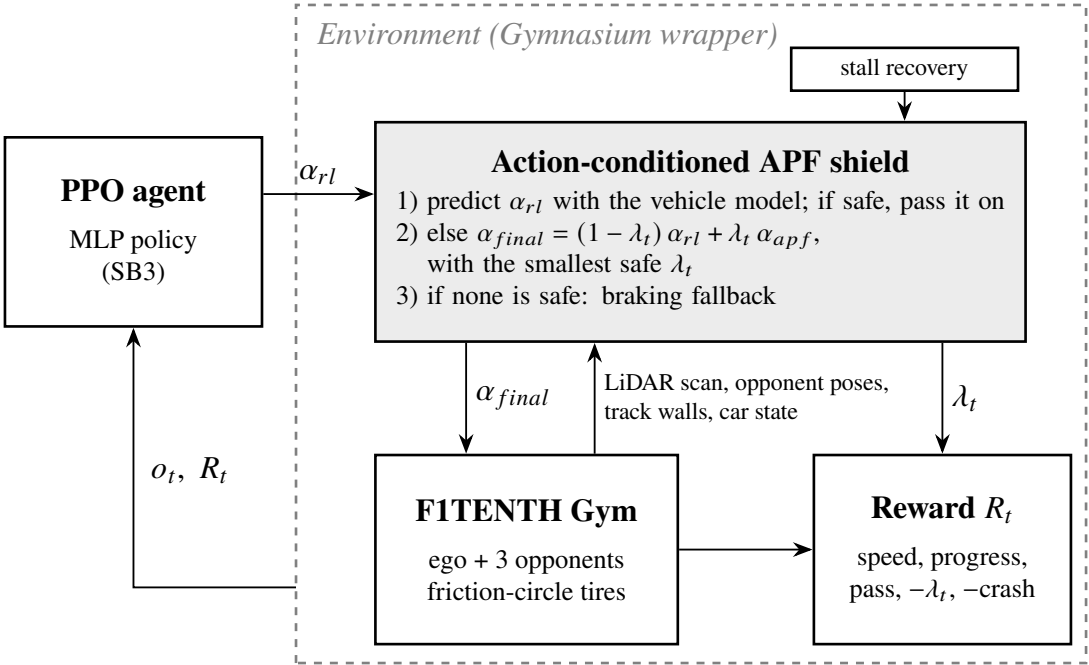
Environment (Gymnasium wrapper)
stall recovery
PPO agent
MLP policy
(SB3)
α_{rl}
Action-conditioned APF shield
1) predict α_{rl} with the vehicle model; if safe, pass it on
2) else $\alpha_{final} = (1 - \lambda_t)\,\alpha_{rl} + \lambda_t\,\alpha_{apf}$,
with the smallest safe λ_t
3) if none is safe: braking fallback
α_{final}
LiDAR scan, opponent poses,
track walls, car state
λ_t
$o_t,\ R_t$
F1TENTH Gym
ego + 3 opponents
friction-circle tires
Reward R_t
speed, progress,
pass, $-\lambda_t$, −crash